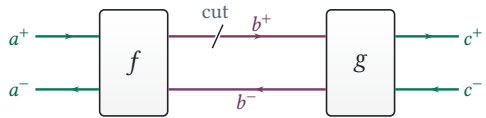
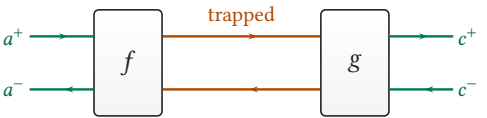
(a) $g \circ f$: a loop over the middle wires



each round trip from f to g and back
multiplies by $f_{12}\, g_{21}$; the composite adds them

(b) two perfect mirrors: $f_{12}\, g_{21} = I$



the light between them is a dead part;
$g \circ f$ is the outer reflection of each mirror